Arc Length and Improper Integrals

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Arc length**

Find the arc length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.

$y' = x^{1/2}$, so $L = \int_0^3 \sqrt{1 + x}\, dx = \left[\frac{2}{3}(1 + x)^{3/2}\right]_0^3 = \frac{2}{3}(8 - 1) = \frac{14}{3}$.

2 **[Foundational]** • **Target: Arc length**

Set up (do not evaluate) the arc length integral for $y = \sin x$ on $[0, \pi]$.

$L = \int_0^{\pi} \sqrt{1 + \cos^2 x}\, dx$.

3 **[Foundational]** • **Target: Improper integrals**

Evaluate $\int_1^{\infty} \frac{1}{x^2}\, dx$, or state that it diverges.

$\lim_{b \to \infty} \left[-\frac{1}{x}\right]_1^b = \lim_{b \to \infty} \left(1 - \frac{1}{b}\right) = 1$: converges to 1.

4 **[Foundational]** • **Target: Improper integrals**

Evaluate $\int_1^{\infty} \frac{1}{x}\, dx$, or state that it diverges.

$\lim_{b \to \infty} [\ln x]_1^b = \lim_{b \to \infty} \ln b = \infty$: diverges.

5 **[Foundational]** • **Target: Improper integrals**

Evaluate $\int_0^1 \frac{1}{\sqrt{x}}\, dx$ (improper at the lower limit).

$\lim_{a \to 0^+} [2\sqrt{x}]_a^1 = 2 - 0 = 2$: converges to 2.

6 **[Foundational]** • **Target: More arc length practice**

Find the arc length of the line $y = 3x + 1$ from $x = 0$ to $x = 4$, and verify your answer using the distance formula.

$y' = 3$: $L = \int_0^4 \sqrt{1 + 9}\, dx = 4\sqrt{10}$. Distance check: endpoints $(0, 1)$ and $(4, 13)$: $\sqrt{4^2 + 12^2} = \sqrt{16 + 144} = \sqrt{160} = 4\sqrt{10}$. ✓

7 **[Foundational]** • **Target: More arc length practice**

Set up (do not evaluate) the arc length integral for $y = \ln x$ on $[1, e]$.

$y' = \frac{1}{x}$: $L = \int_1^e \sqrt{1 + \frac{1}{x^2}}\, dx$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: More improper integrals

Evaluate $\int_1^\infty \frac{1}{x^{1.5}}\,dx$, or state that it diverges.

$\lim_{b\to\infty}[-2x^{-0.5}]_1^b = \lim_{b\to\infty}\left(2-\frac{2}{\sqrt{b}}\right) = 2$: converges to 2.

9 [Intermediate] • Target: More improper integrals

Evaluate $\int_{-\infty}^0 e^x\,dx$, or state that it diverges.

$\lim_{a\to-\infty}[e^x]_a^0 = 1-\lim_{a\to-\infty} e^a = 1-0 = 1$: converges to 1.

10 [Intermediate] • Target: The p-integral test

For $\int_1^\infty \frac{1}{x^p}\,dx$, state the condition on p for convergence, and explain why using the antiderivative $\frac{x^{1-p}}{1-p}$ (for $p\neq 1$).

Converges when $p>1$. As $b\to\infty$, $b^{1-p}\to 0$ only when $1-p<0$, i.e. $p>1$; otherwise the limit is infinite (or, for $p=1$, the antiderivative is $\ln x$, which also diverges).

11 [Intermediate] • Target: The p-integral test

Use the p-integral result to quickly determine whether $\int_1^\infty \frac{1}{x^{0.9}}\,dx$ converges.

$p=0.9<1$: diverges.

12 [Intermediate] • Target: Improper integrals with infinite discontinuities inside the interval

Evaluate $\int_0^2 \frac{1}{(x-1)^2}\,dx$, or explain why it diverges (note the integrand is undefined at $x=1$, inside the interval).

Split at $x=1$: $\int_0^1 \frac{1}{(x-1)^2}dx + \int_1^2 \frac{1}{(x-1)^2}dx$. Near $x=1$ from either side, $\lim_{b\to 1}\left[-\frac{1}{x-1}\right]$ blows up to $\pm\infty$: both pieces diverge, so the full integral diverges.

13 [Intermediate] • Target: Improper integrals with infinite discontinuities inside the interval

Evaluate $\int_{-1}^1 \frac{1}{x^{1/3}}\,dx$ (an odd function with a discontinuity at $x=0$), splitting at $x=0$.

By symmetry (odd integrand), $\int_{-1}^0 x^{-1/3}dx = -\int_0^1 x^{-1/3}dx$ (each piece converges since $p=\frac{1}{3}<1$ for this type of integrand near a finite discontinuity). The two pieces cancel: the integral equals 0.

14 [Intermediate] • Target: Visualizing an improper integral as a limiting area

The graph shows $f(x)=\frac{1}{x^2}$ on $[1,10]$. As the right boundary extends to infinity, the shaded area approaches a finite limit. Explain why this is plausible, using the fact that $\int_1^\infty \frac{1}{x^2}\,dx = 1$ (found earlier).

The curve $y=\frac{1}{x^2}$ decreases rapidly toward 0, so the strips added as x grows contribute less and less area. Even though the region extends infinitely far right, the total shaded area converges to a finite value, 1, because the tail contributions shrink fast enough (consistent with the p-test, $p=2>1$).

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Arc length of a more involved curve** • **Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals**

Set up (do not evaluate) the arc length integral for $y = x^2$ on $[0, 2]$, and estimate it using the straight-line distance between the endpoints as a lower bound.

$y' = 2x$: $L = \int_0^2 \sqrt{1 + 4x^2}\, dx$. Straight-line distance between $(0, 0)$ and $(2, 4)$: $\sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5} \approx 4.47$, which is a lower bound since the actual curve is longer than the chord connecting its endpoints.

16 **[Challenging]** • **Target: Comparison test for improper integrals** • **Cross-Lesson: 4.02 Riemann Sums and Definite Integrals**

Use a comparison with $\int_1^\infty \frac{1}{x^2}\, dx$ (known to converge) to argue that $\int_1^\infty \frac{1}{x^2 + 1}\, dx$ also converges, without evaluating it.

For $x \geq 1$, $\frac{1}{x^2 + 1} < \frac{1}{x^2}$. Since $\int_1^\infty \frac{1}{x^2}\, dx$ converges (to 1), and the integrand of interest is smaller and positive everywhere on $[1, \infty)$, the comparison test guarantees $\int_1^\infty \frac{1}{x^2 + 1}\, dx$ also converges.

17 **[Challenging]** • **Target: Comparison test for improper integrals** • **Cross-Lesson: 5.01 The Fundamental Theorem of Calculus**

Use a comparison with $\int_1^\infty \frac{1}{x}\, dx$ (known to diverge) to argue that $\int_1^\infty \frac{1}{\sqrt{x}}\, dx$ also diverges.

For $x \geq 1$, $\frac{1}{\sqrt{x}} \geq \frac{1}{x}$ (since $\sqrt{x} \leq x$). Since $\int_1^\infty \frac{1}{x}\, dx$ diverges and the integrand of interest is even larger, the comparison test guarantees $\int_1^\infty \frac{1}{\sqrt{x}}\, dx$ also diverges.

18 **[Challenging]** • **Target: A capstone arc-length application** • **Cross-Lesson: 6.01 Related Rates**

A cable hangs in the shape $y = \frac{x^2}{8}$ between the towers at $x = -4$ and $x = 4$. Set up the integral for the length of the cable (do not evaluate).

$y' = \frac{x}{4}$: $L = \int_{-4}^4 \sqrt{1 + \frac{x^2}{16}}\, dx$.

19 **[Challenging]** • **Target: A capstone arc-length application** • **Cross-Lesson: 6.02 Optimization and Curve Sketching**

Evaluate $\int_2^\infty \frac{1}{(x-1)^3}\, dx$, or state that it diverges.

$\lim_{b \to \infty} \left[-\frac{1}{2(x-1)^2}\right]_2^b = \lim_{b \to \infty} \left(-\frac{1}{2(b-1)^2} + \frac{1}{2}\right) = \frac{1}{2}$: converges to $\frac{1}{2}$.

20 **[Challenging]** • **Target: A capstone arc-length application** • **Cross-Lesson: 7.01 Integration by Substitution**

A city's water tower is shaped by revolving the curve $y = \sqrt{25 - x^2}$ (the upper half of a circle of radius 5) about the x-axis; set up (do not evaluate) the arc length integral for this curve on $[-4, 4]$, which describes the profile of one side of the tower.

$y' = \frac{-x}{\sqrt{25 - x^2}}$: $L = \int_{-4}^4 \sqrt{1 + \frac{x^2}{25 - x^2}}\, dx = \int_{-4}^4 \sqrt{\frac{25}{25 - x^2}}\, dx = \int_{-4}^4 \frac{5}{\sqrt{25 - x^2}}\, dx$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Arc Length of a Plane Curve

State the formula for the arc length L of $y = f(x)$ from $x = a$ to $x = b$.

a. $L = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx$ ✓

b. $L = \int_a^b \sqrt{1 + f'(x)}\, dx$

c. $L = \int_a^b [1 + f'(x)]\, dx$

d. $L = \int_a^b \sqrt{[f(x)]^2 + [f'(x)]^2}\, dx$

From $ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + (dy/dx)^2}\, dx$, arc length is $\int_a^b \sqrt{1 + [f'(x)]^2}\, dx$.

22 [Foundational] • Target: Arc Length of a Plane Curve

Find the arc length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.

a. $\frac{14}{3}$ ✓

b. 7

c. $\frac{16}{3}$

d. 8

$y' = x^{1/2} \Rightarrow 1 + (y')^2 = 1 + x$. $L = \int_0^3 \sqrt{1 + x}\, dx = [\frac{2}{3}(1 + x)^{3/2}]_0^3 = \frac{2}{3}(4^{3/2} - 1^{3/2}) = \frac{2}{3}(8 - 1) = \frac{14}{3}$.

23 [Foundational] • Target: Arc Length of a Plane Curve

Find the arc length of the line segment $y = \frac{4}{3}x + 2$ from $x = 0$ to $x = 3$.

a. 5 ✓

b. 4

c. 6

d. $\sqrt{7}$

$y' = 4/3 \Rightarrow \sqrt{1 + 16/9} = \sqrt{25/9} = 5/3$. $L = \int_0^3 \frac{5}{3}\, dx = \frac{5}{3}(3) = 5$.

24 [Foundational] • Target: Improper Integrals of Type I (Infinite Bounds)

Evaluate $\int_1^\infty \frac{1}{x^2}\, dx$.

a. 1 ✓

b. ∞ (diverges)

c. 1/2

d. 2

$\lim_{b\to\infty} \int_1^b x^{-2}\, dx = \lim_{b\to\infty} [-1/x]_1^b = \lim_{b\to\infty} (-1/b + 1) = 1$.

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25 [Foundational] • Target: Improper Integrals of Type I (Infinite Bounds)

Evaluate $\int_0^\infty e^{-2x}\,dx$.

a. $1/2$ ✓ **b.** 1

c. ∞ **d.** 2

$\lim_{b\to\infty}[-\frac{1}{2}e^{-2x}]_0^b = \lim_{b\to\infty}(-\frac{1}{2}e^{-2b}+1/2) = 1/2.$

26 [Foundational] • Target: Improper Integrals of Type I (Infinite Bounds)

For what values of p does the improper integral $\int_1^\infty \frac{1}{x^p}\,dx$ converge?

a. $p>1$ ✓ **b.** $p\geq 1$

c. $p<1$ **d.** $p>0$

By the p-integral test on $[1,\infty)$, $\int_1^\infty x^{-p}\,dx$ converges if and only if $p>1$. For $p\leq 1$, it diverges.

27 [Foundational] • Target: Improper Integrals of Type II (Discontinuous Integrands)

Evaluate $\int_0^1 \frac{1}{\sqrt{x}}\,dx$.

a. 2 ✓ **b.** 1

c. ∞ (diverges) **d.** $1/2$

Integrand undefined at $x=0$. $\lim_{a\to 0^+}[2\sqrt{x}]_a^1 = \lim_{a\to 0^+}(2-2\sqrt{a}) = 2.$

28 [Foundational] • Target: Improper Integrals of Type II (Discontinuous Integrands)

Evaluate $\int_0^2 \frac{1}{x-1}\,dx$.

a. **Diverges** ✓ **b.** 0

c. $\ln 2$ **d.** ∞

Infinite discontinuity at $x=1$. Split: $\int_0^1 + \int_1^2$. $\int_0^1 \frac{dx}{x-1} = \lim_{b\to 1^-}[\ln|x-1|]_0^b = \lim_{b\to 1^-}\ln|b-1| - 0 = -\infty$. Since one part diverges, the integral diverges.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Improper Integrals of Type II (Discontinuous Integrands)

For what values of p does $\int_0^1 \frac{1}{x^p}\,dx$ converge?

a. $p < 1$ ✓

b. $p > 1$

c. $p \le 1$

d. $p > 0$

Near 0, $\int_0^1 x^{-p}\,dx = [\frac{x^{1-p}}{1-p}]_0^1$. This converges if and only if the exponent $1-p > 0 \Rightarrow p < 1$.

30 [Intermediate] • Target: Comparison Test for Improper Integrals

Determine the convergence of $\int_1^\infty \frac{\sin^2(x)}{x^2}\,dx$.

a. **Converges by comparison with $\frac{1}{x^2}$** ✓

b. Diverges

c. Oscillates without limit

d. Converges to 0

Since $0 \le \frac{\sin^2(x)}{x^2} \le \frac{1}{x^2}$ and $\int_1^\infty \frac{1}{x^2}\,dx$ converges, by the Direct Comparison Test the integral converges.

31 [Intermediate] • Target: Comparison Test for Improper Integrals

Determine the convergence of $\int_1^\infty \frac{1}{\sqrt{x^3+1}}\,dx$.

a. **Converges (behaves like $x^{-3/2}$ as $x \to \infty$)** ✓

b. Diverges (behaves like $x^{-1/2}$)

c. Diverges to $-\infty$

d. Cannot be tested

For large x, $\frac{1}{\sqrt{x^3+1}} \approx \frac{1}{x^{3/2}}$. By the Limit Comparison Test with $p = 3/2 > 1$, the integral converges.

32 [Intermediate] • Target: Comparison Test for Improper Integrals

Evaluate $\int_{-\infty}^\infty \frac{1}{1+x^2}\,dx$.

a. π ✓

b. 2π

c. $\pi/2$

d. Diverges

Split at 0: $2\int_0^\infty \frac{1}{1+x^2}\,dx = 2\lim_{b\to\infty}[\arctan x]_0^b = 2(\pi/2 - 0) = \pi$.

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33 [Intermediate] • Target: Improper Integrals of Type II (Discontinuous Integrands)

Comparison test for improper integrals: $0 \le f(x) \le g(x)$, $\int_a^\infty g$ converges implies:

a. $\int_a^\infty f$ **converges** ✓

b. $\int_a^\infty f$ diverges

c. $f = g$

d. nothing

Comparison: dominated convergent integral converges.

34 [Intermediate] • Target: Comparison Test for Improper Integrals

Evaluate the improper integral $\int_0^\infty \frac{1}{x^2+9}\,dx$.

a. $\frac{\pi}{3}$

b. ∞

c. $\frac{\pi}{6} + 1$

d. $\frac{\pi}{6}$ ✓

Write it as a limit, e.g. $\lim_{b\to\infty} \int_0^b \frac{1}{x^2+9}\,dx$, integrate, and take the limit. The integral converges to $\frac{\pi}{6}$.

35 [Intermediate] • Target: Arc length

Evaluate the improper integral $\int_0^\infty x e^{-3x^2}\,dx$.

a. $\frac{1}{3}$

b. ∞

c. $\frac{1}{6}$ ✓

d. $\frac{7}{6}$

Write it as a limit, e.g. $\lim_{b\to\infty} \int_0^b x e^{-3x^2}\,dx$, integrate, and take the limit. The integral converges to $\frac{1}{6}$.

36 [Intermediate] • Target: Improper integrals

Improper integral: $\int_1^\infty \frac{1}{x^2}\,dx =$

a. 1 ✓

b. ∞

c. 1/2

d. 2

Applying Integrals principles, $[-1/x]_1^\infty = 0 - (-1) = 1$. Therefore, the verified correct option is 1.

37 [Intermediate] • Target: More arc length practice

Improper double integral: $\int_0^\infty \int_0^\infty e^{-(x+y)}dx\,dy =$

a. 1 ✓

b. 0

c. ∞

d. e

$= \left(\int_0^\infty e^{-x}dx\right)^2 = 1^2 = 1.$

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: More improper integrals • Cross-Lesson: 6.01 Related Rates

Integral test: $\sum^{\infty} f(n)$ converges iff $\int_1^{\infty} f(x)dx$ converges, for f positive decreasing. So $\sum 1/n^2$ converges because $\int_1^{\infty} x^{-2}dx =$

a. 1 **(finite)** ✓

b. ∞

c. 0

d. $1/2$

$\int_1^{\infty} x^{-2}dx = 1$: finite.

39 [Challenging] • Target: The p-integral test • Cross-Lesson: 6.02 Optimization and Curve Sketching

Find the area inside $r = 10\cos(\theta)$ and outside $r = 5$.

a. 25π

b. $25\sqrt{3} + \frac{50\pi}{3}$

c. $\frac{25\pi}{3}$

d. $\frac{25\sqrt{3}}{2} + \frac{25\pi}{3}$ ✓

The curves meet at $\theta = \pm\pi/3$. $A = \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \left(50\cos^2(\theta) - \frac{25}{2}\right) d\theta = \frac{25\sqrt{3}}{2} + \frac{25\pi}{3}$.

40 [Challenging] • Target: Improper integrals with infinite discontinuities inside the interval • Cross-Lesson: 7.01 Integration by Substitution

Find the area inside $r = 2\cos(\theta)$ and outside $r = 1$.

a. π

b. $\sqrt{3} + \frac{2\pi}{3}$

c. $\frac{\sqrt{3}}{2} + \frac{\pi}{3}$ ✓

d. $\frac{\pi}{3}$

The curves meet at $\theta = \pm\pi/3$. $A = \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \left(2\cos^2(\theta) - \frac{1}{2}\right) d\theta = \frac{\sqrt{3}}{2} + \frac{\pi}{3}$.

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41 [Challenging] • Target: Visualizing an improper integral as a limiting area • Cross-Lesson: 7.02 Integration by Parts

Find the area inside $r = 8\cos(\theta)$ and outside $r = 4$.

a. 16π

b. $16\sqrt{3} + \frac{32\pi}{3}$

c. $\frac{16\pi}{3}$

d. $8\sqrt{3} + \frac{16\pi}{3}$ ✓

The curves meet at $\theta = \pm\pi/3$. $A = \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} (32\cos^2(\theta) - 8)\, d\theta \;=\; 8\sqrt{3} + \frac{16\pi}{3}$.

42 [Challenging] • Target: Arc length of a more involved curve • Cross-Lesson: 8.01 Area Between Curves

Find the area inside $r = 4\cos(\theta)$ and outside $r = 2$.

a. 4π

b. $2\sqrt{3} + \frac{4\pi}{3}$ ✓

c. $4\sqrt{3} + \frac{8\pi}{3}$

d. $\frac{4\pi}{3}$

The curves meet at $\theta = \pm\pi/3$. $A = \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} (8\cos^2(\theta) - 2)\, d\theta \;=\; 2\sqrt{3} + \frac{4\pi}{3}$.

43 [Challenging] • Target: Comparison test for improper integrals • Cross-Lesson: 8.02 Volumes of Revolution

Find the arc length of $y = \cosh(x)$ from $x = 0$ to $x = \ln(3)$.

a. $\frac{4}{3}$ ✓

b. $\frac{2}{3}$

c. $\frac{7}{3}$

d. $\frac{8}{3}$

$L = \int_0^{\ln(3)} \sqrt{y'^2 + 1}\, dx$; here $y'^2 + 1 = \cosh^2(x)$ is a perfect square, so the integral evaluates to $\frac{4}{3}$.

44 [Challenging] • Target: A capstone arc-length application • Cross-Lesson: 9.01 Separable Differential Equations

Find the arc length of $y = \frac{x^3}{6} + \frac{1}{2x}$ from $x = 1$ to $x = 3$.

a. 4

b. $\frac{17}{3}$

c. $\frac{28}{3}$

d. $\frac{14}{3}$ ✓

$L = \int_1^3 \sqrt{y'^2 + 1}\, dx$; here $y'^2 + 1 = \frac{(x^4+1)^2}{4x^4}$ is a perfect square, so the integral evaluates to $\frac{14}{3}$.

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45 [Challenging] • Target: Arc Length of a Plane Curve • Cross-Lesson: 1.01 Precalculus Review for Calculus

The improper integral $\int_0^1 \ln x \, dx$:

a. converges to -1 ✓

b. diverges to $-\infty$

c. converges to 0

d. converges to 1

Antiderivative $x\ln x - x$; as $x \to 0^+$, $x\ln x \to 0$, so the value is $(1 \cdot 0 - 1) - (0 - 0) = -1$, a finite limit despite the unbounded integrand near 0.

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